

Q1. Solution:

Since, $\frac{d}{dx}(y^3) = 3y^2 y'$

We have $y^2 y' = \frac{1}{3} \frac{d}{dx}(y^3)$

Substituting in the given differential equation:

$xy^2 y' + y^3 = \frac{\sin x}{x}$, we get

$$\frac{x}{3} \frac{d}{dx}(y^3) + y^3 = \frac{\sin x}{x}$$

$$\Rightarrow x \frac{d}{dx}(y^3) + 3y^3 = \frac{3 \sin x}{x} \quad \text{--- (*)}$$

Let $u = y^3$.

Then, $\frac{du}{dx} = \frac{d}{dx}(y^3)$

From (*) we get:

$$x \frac{du}{dx} + 3u = \frac{3 \sin x}{x}$$

$$\text{or } x u' + 3u = \frac{3 \sin x}{x}$$

2

$$\Rightarrow u' + \frac{3}{x}u = \frac{3\sin x}{x^2}$$

This is a linear differential equation.

So, the integrating factor, denoted by

I.F. is:

$$\text{I.F.} = e^{\int \frac{3}{x} dx} = e^{3 \ln x} = x^3.$$

So,

$$x^3 u = \int \frac{3 \sin x}{x^2} \cdot x^3 dx + C$$

$$\Rightarrow x^3 u = 3 \int x \sin x dx + C$$

$$= 3 [-x \cos x + \int \cos x dx] + C$$

$$= 3 [-x \cos x + \sin x] + C$$

$$\Rightarrow u = \frac{3}{x^3} [-x \cos x + \sin x] + \frac{C}{x^3}$$

Substitute $u = y^3$, we get

$$y^3 = \frac{-3x \cos x + 3 \sin x + C}{x^3}$$

→ General solⁿ.

Now; $y\left(\frac{\pi}{2}\right) = 0$

$$\Rightarrow 0^3 = \frac{-3\left(\frac{\pi}{2}\right) \cos\left(\frac{\pi}{2}\right) + 3 \sin\left(\frac{\pi}{2}\right) + C}{\left(\frac{\pi}{2}\right)^3}$$

$$\Rightarrow 0 = \frac{3+C}{\left(\frac{\pi}{2}\right)^3}$$

$$\Rightarrow C = -3$$

Particular solution!!

Hence, $y^3 = \frac{-3x \cos x + 3 \sin x - 3}{x^3}$

At $x = \frac{5\pi}{2}$, $\sin\left(\frac{5\pi}{2}\right) = 1$
 $\cos\left(\frac{5\pi}{2}\right) = 0$

So,

$$y^3 = \frac{-3(0) + 3(1) - 3}{\left(\frac{5\pi}{2}\right)^3} \Rightarrow y = 0 \text{ at } x = \frac{5\pi}{2}$$

ANSWER !!

Alternative way which students can use to reach to a General Solution.

Note that: $\frac{d}{dx}(y^3) = 3y^2 y'$

The given Equation is:

$$xy^2 y' + y^3 = \frac{\sin x}{x}$$

$$\Rightarrow 3xy^2 y' + 3y^3 = \frac{3 \sin x}{x}$$

$$\textcircled{3} \Rightarrow x \frac{d}{dx}(y^3) + 3y^3 = \frac{3 \sin x}{x}$$

$$\Rightarrow x^3 \frac{d}{dx}(y^3) + 3x^2 y^3 = 3x^2 \frac{\sin x}{x}$$

$$\Rightarrow x^3 \frac{d}{dx}(y^3) + 3x^2 y^3 = 3x \sin x$$

$$\Rightarrow \frac{d}{dx}(x^3 y^3) = 3x \sin x$$

on integrating:

$$\Rightarrow x^3 y^3 = \int 3x \sin x dx + C$$

$$\textcircled{3} \Rightarrow x^3 y^3 = 3 [-x \cos x + \sin x] + C$$

$$\Rightarrow y^3 = \frac{3 [-x \cos x + \sin x] + C}{x^3}$$

↓ They'll proceed in finding the particular solⁿ. as above.

Another way

Given eqⁿ: $xy^2y' + y^3 = \frac{\sin x}{x}$

$$\Rightarrow xy^2 dy + y^3 dx = \frac{\sin x}{x} dx$$

$$\textcircled{2} \Rightarrow \left(y^3 - \frac{\sin x}{x} \right) dx + xy^2 dy = 0$$

$$\text{Let } M = y^3 - \frac{\sin x}{x}, \quad N = xy^2$$

$$\text{Then, } \frac{\partial M}{\partial y} = 3y^2, \quad \frac{\partial N}{\partial x} = y^2$$

$$\begin{aligned} \text{So, } \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) &= \frac{1}{xy^2} (3y^2 - y^2) \\ &= \frac{2y^2}{xy^2} \\ &= \frac{2}{x} \quad (\text{function of } x \text{ only}) \end{aligned}$$

$$\textcircled{1} \text{ So, I.F} = e^{\int \frac{2}{x} dx} = e^{2 \ln x} = x^2.$$

Multiply the given differential eqⁿ by x^2 , we get:

$$(x^2 y^3 - x \sin x) dx + x^3 y^2 dy = 0$$

This is an exact differential eqⁿ.

So, $\int (x^2 y^3 - x \sin x) dx$

||

③ $\frac{x^3}{3} y^3 + x \cos x - \sin x + g(y)$, (treating 'y' as constant)

on differentiating w.r.t 'y' we get:

$$x^3 y^2 + g'(y)$$

So, on comparing we get: $x^3 y^2 + g'(y) = x^3 y^2$

$$\Rightarrow g'(y) = 0 \Rightarrow g(y) = \text{constant.}$$

So, General solution is:

$$\frac{x^3 y^3}{3} + x \cos x - \sin x = C$$

$$\Rightarrow x^3 y^3 + 3x \cos x - 3 \sin x = C$$

$$\Rightarrow y^3 = \frac{3 [-x \cos x + \sin x] + C}{x^3}$$

Remaining part is as shown before.

Q2. Solution

$$D = \{ (x, y) \in \mathbb{R}^2 \mid -1 \leq x \leq 1, -1 \leq y \leq 1 \}$$

$$f(x, y) = 1 + \sqrt{\hat{y}}, \text{ where } \hat{y} = \max\{y, 0\}$$

and the Ivp is

$$y' = f(x, y), \quad y(0) = 0.$$

We know that

a) ↑ 'f' satisfies Lipschitz condⁿ on D if

∃ M > 0 st.

$$|f(x, y_1) - f(x, y_2)| \leq M |y_1 - y_2| \dots (1)$$

, for every $(x, y_1), (x, y_2) \in D$.

Now, f can be written piecewise as

$$f(x, y) = \begin{cases} 1 & ; y \leq 0 \\ 1 + \sqrt{y} & ; y > 0 \end{cases}$$

as,

$$\hat{y} = \begin{cases} 0 & ; y \leq 0 \\ y & ; y > 0 \end{cases}$$

Take $y_1 = 0$ $y_2 = \epsilon > 0$ then $\hat{y}_1 = 0$ & $\hat{y}_2 = \epsilon$

Then $f(x, y_1) = f(x, 0) = 1$ and

$$f(x, y_2) = f(x, \epsilon) = 1 + \sqrt{\epsilon}$$

$$\text{so, } |f(x, 0) - f(x, \epsilon)| = |1 - (1 + \sqrt{\epsilon})| = \sqrt{\epsilon}$$

$$\text{and } |y_1 - y_2| = \epsilon$$

So, if f satisfies Lipschitz condⁿ w.r.t 'y'
then

$$|f(x, y_1) - f(x, y_2)| \leq M |y_1 - y_2|$$

$$\Rightarrow \sqrt{\epsilon} \leq M \epsilon$$

$$\Rightarrow \frac{1}{\sqrt{\epsilon}} \leq M \quad \text{---} (*)$$

$$\text{As } \epsilon \rightarrow 0^+$$

$$\Rightarrow \frac{1}{\sqrt{\epsilon}} \rightarrow \infty$$

3 So, no finite M can satisfy the inequality (*) as $\epsilon > 0$ is arbitrary and $\frac{1}{\sqrt{\epsilon}}$ can be made as large as we please.

$\Rightarrow$ f doesn't satisfy Lipschitz condⁿ on D .

(b) i) Checking continuity of $f(x,y)$ on D .

Case-1

when $y < 0$,

$$f(x,y) = 1$$

i.e, f is a constant function for all $(x,y) \in D$ with $y < 0$ hence is continuous.

Case-2

when $y > 0$ then

$$f(x,y) = 1 + \sqrt{y}$$

$\Rightarrow 1 + \sqrt{y} = f(x,y)$, for all $(x,y) \in D$ with $y > 0$ and hence is continuous.

Case-3

when $y = 0$:

observe that $\lim_{y \rightarrow 0^-} f(x,y) = 1$

②

$$\text{and } \lim_{y \rightarrow 0^+} f(x,y) = 1 + \sqrt{0} = 1$$

$$\text{and } f(x,0) = 1.$$

$\Rightarrow f(x,y)$ is continuous for all $(x,y) \in D$ with $y = 0$.

$\Rightarrow$ f is continuous on D .

(iii) Checking boundedness of $f(x,y)$ on D :

As on D , $-1 \leq y \leq 1$.

Observe that,

when $y \leq 0$ then $f(x,y) = 1$.

when $0 \leq y \leq 1$ then $f(x,y) = 1 + \sqrt{y}$

Now, on $0 \leq y \leq 1$,

$$0 \leq \sqrt{y} \leq 1$$

②

$$\Rightarrow 1 \leq 1 + \sqrt{y} \leq 2$$

$\therefore$ for all $(x,y) \in D$,

$$1 \leq f(x,y) \leq 2$$

In particular, $|f(x,y)| \leq 2$, $\forall (x,y) \in D$.

$\Rightarrow$ f is bounded on D .

$\therefore$ f is continuous on D and is bounded on D , then by existence theorem the IVP $y' = f(x,y)$ with $y(0) = 0$

has atleast one solution.

①

3. Solution:

Since, the given differential equation is exact on D .

There exists a differentiable function $F(x,y)$ on D such that $dF = M(x,y)dx + N(x,y)dy = 0$.

③ $\Rightarrow \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = M(x,y)dx + N(x,y)dy$

on comparing, we get $\frac{\partial F}{\partial x} = M(x,y)$ and $\frac{\partial F}{\partial y} = N(x,y)$.

so, $\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial x} \right) = \frac{\partial^2 F}{\partial y \partial x}$

② and $\frac{\partial N}{\partial x} = \frac{\partial}{\partial x} \left(\frac{\partial F}{\partial y} \right) = \frac{\partial^2 F}{\partial x \partial y}$

Since F is a differentiable function $\Rightarrow$ it has continuous first order partial derivatives.

② $S_0,$

$$\frac{\partial^2 F(x,y)}{\partial y \partial x} = \frac{\partial^2 F(x,y)}{\partial x \partial y}$$

①

$$\therefore \frac{\partial M(x,y)}{\partial y} = \frac{\partial N(x,y)}{\partial x}, \forall (x,y) \in D.$$

Note: If they have shown the essential logic, please grade leniently.

④ Solution:

Given, $\frac{dy_1}{dx} = 5y_1 - 2y_2$

$$\frac{dy_2}{dx} = 4y_1 - y_2 \text{ with the}$$

initial condⁿs:

$$y_1(0) = 0$$

$$y_2(0) = 1$$

a) We can write the above system equivalently as:

$$\begin{bmatrix} \frac{dy_1}{dx} \\ \frac{dy_2}{dx} \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

or $Y' = AY$, where $Y' = \begin{bmatrix} \frac{dy_1}{dx} \\ \frac{dy_2}{dx} \end{bmatrix}$,

$$A = \begin{bmatrix} 5 & -2 \\ 4 & -1 \end{bmatrix} \text{ \& } Y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

We'll now calculate the eigenvalue of A:

$$\text{det}(A - \lambda I) = \begin{vmatrix} 5-\lambda & -2 \\ 4 & -1-\lambda \end{vmatrix} = 0$$

②

$$\Rightarrow (5-\lambda)(-1-\lambda) + 8 = 0 \quad (\text{characteristic eqn})$$

$$\Rightarrow \lambda^2 - 4\lambda + 3 = 0$$

$$\Rightarrow (\lambda - 3)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = 3 \text{ and } 1.$$

• Eigenvector corresponding to $\lambda = 3$:

Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be an eigenvector corresponding to $\lambda = 3$. then

$$(A - 3I)\vec{x} = 0$$

$$\Rightarrow \begin{bmatrix} 2 & -2 \\ 4 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow 2x_1 - 2x_2 = 0$$

$$4x_1 - 4x_2 = 0$$

$$\Rightarrow x_1 = x_2.$$

So, $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector
① corresponding to an eigenvalue $\lambda = 3$.

• Eigenvector corresponding to an eigenvalue $\lambda = 1$

We again solve

$$(A - 1 \cdot I) \vec{x} = 0 \text{ for } \vec{x}.$$

$$\therefore (A - I) \vec{x} = \begin{bmatrix} 4 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

① $\Rightarrow 4x_1 - 2x_2 = 0$

$$\Rightarrow x_2 = 2x_1$$

So, an eigenvector corresponding to an eigenvalue $\lambda = 1$ is

$$\vec{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$\therefore$ the general solⁿ of the system is

$$\vec{y}(x) = c_1 e^{\lambda_1 x} \vec{v}_1 + c_2 e^{\lambda_2 x} \vec{v}_2$$

$$= c_1 e^{3x} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^x \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

So,

$$\begin{bmatrix} y_1(x) \\ y_2(x) \end{bmatrix} = \begin{bmatrix} c_1 e^{3x} + c_2 e^x \\ c_1 e^{3x} + 2c_2 e^x \end{bmatrix} \quad \text{where } c_1 \text{ and } c_2 \text{ are arbitrary constants.}$$

on applying initial condⁿs

$$y_1(0) = 0$$

$$y_2(0) = 1, \text{ we have}$$

$$\begin{bmatrix} y_1(0) \\ y_2(0) \end{bmatrix} = \begin{bmatrix} c_1 + c_2 \\ c_1 + 2c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow c_1 + c_2 = 0$$

$$\& c_1 + 2c_2 = 1$$

$$\Rightarrow c_1 = -1 \text{ and } c_2 = 1$$

Solution for the system of equation

$$\begin{aligned} y_1(x) &= -e^{3x} + e^x \\ y_2(x) &= -e^{3x} + 2e^x \end{aligned}$$

$$\text{OR} \quad \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = - \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^x$$

(b)

For the critical point:

$$\frac{dy_1}{dx} = 0, \quad \frac{dy_2}{dx} = 0$$

So,

$$5y_1 - 2y_2 = 0$$

$$4y_1 - y_2 = 0 \times 2$$

$$\begin{array}{rcl} \text{we get,} & 5y_1 - 2y_2 = 0 & \\ & -8y_1 + 2y_2 = 0 & \\ \hline & -3y_1 = 0 & \end{array}$$

$$\Rightarrow y_1 = 0 \text{ and } y_2 = 0.$$

So,

critical point : $(y_1, y_2) = (0, 0)$

c) For the matrix A, eigenvalues are 1 & 3.
Since eigenvalues of A are real and of
Same Sign (or since $\det A = \lambda_1 \lambda_2 > 0$
and $\Delta = (\lambda_1 - \lambda_2)^2 > 0$),

1+1
(0,0) is a node.

Since both the eigenvalues are positive, $\text{tr}(A) > 0$
and so (0,0) is an unstable critical point. ■

Q5 Solution:

Given that f_1 and f_2 are two solutions.

$$\text{So, } \forall x \in [a, b], \quad f_1''(x) + a_1(x) f_1'(x) + a_2(x) f_1(x) = 0$$
$$\textcircled{1} \quad f_2''(x) + a_1(x) f_2'(x) + a_2(x) f_2(x) = 0$$

Substituting $x = x_0$,

$$\textcircled{2} \quad a_1(x_0) f_1'(x_0) + a_2(x_0) f_1(x_0) = 0$$

and $a_1(x_0) f_2'(x_0) + a_2(x_0) f_2(x_0) = 0$

$$\Rightarrow \begin{bmatrix} f_1'(x_0) & f_1(x_0) \\ f_2'(x_0) & f_2(x_0) \end{bmatrix} \begin{bmatrix} a_1(x_0) \\ a_2(x_0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \dots \textcircled{1}$$

Since f_1 and f_2 are linearly independent on $[a, b]$,

$$\Rightarrow w(f_1, f_2)(x_0) = \begin{vmatrix} f_1(x_0) & f_2(x_0) \\ f_1'(x_0) & f_2'(x_0) \end{vmatrix}$$

$\textcircled{2}$

$$\neq 0$$

$$\Rightarrow f_1(x_0) f_2'(x_0) - f_2(x_0) f_1'(x_0) \neq 0$$

Since $\begin{vmatrix} f_1'(x_0) & f_1(x_0) \\ f_2'(x_0) & f_2(x_0) \end{vmatrix} = f_1'(x_0) f_2(x_0) - f_1(x_0) f_2'(x_0)$

$$= -w(f_1, f_2)(x_0) \neq 0,$$

③

the above system of equation ① has only trivial solution.

Hence $\begin{bmatrix} a_1(x_0) \\ a_2(x_0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow a_1(x_0) = a_2(x_0) = 0.$



Q6: solution

Given differential eqⁿ is

$$\frac{dy}{dx} - 2ye^x = 2e^x \sqrt{y} ; y(0) = 1$$

We know the standard Bernoulli's equation as:

$$\textcircled{2} \quad \frac{dy}{dx} + p(x)y = Q(x)y^n$$

Comparing it with given differential

eqⁿ, we have $p(x) = -2e^x$

$$Q(x) = 2e^x$$

and $n = \frac{1}{2}$

and substitute $V = \sqrt{y}$
 $\Rightarrow V^2 = y$

$$\Rightarrow \frac{dy}{dx} = 2V \frac{dV}{dx}$$

$\textcircled{2}$ Putting in original equation:

$$2V \frac{dV}{dx} - 2e^x V^2 = 2e^x V$$

$$\Rightarrow \frac{dv}{dx} - e^x v = e^x.$$

$$\text{So, I.F} = e^{\int -e^x dx} = e^{-e^x}$$

Hence,

$$v \times \text{I.F} = \int e^x \times \text{I.F} dx$$

$$\Rightarrow v e^{-e^x} = \int e^x \cdot e^{-e^x} dx$$

②

Let $t = -e^x$ then $dt = -e^x dx$.

$$\text{So, } \int e^x e^{-e^x} dx = -\int e^t dt = -e^t + c.$$

$$\Rightarrow v e^{-e^x} = -e^{-e^x} + c$$

$$\Rightarrow v = -1 + c e^{e^x}$$

Again putting $v^2 = y$, we get

$$y = (-1 + c e^{e^x})^2.$$

For initial condⁿ $y(0) = 1$;

we have,

$$(-1+ce)^2 = 1$$

$$\Rightarrow -1+ce = 1 \quad \text{or} \quad -1+ce = -1$$

$$\Rightarrow ce = 2$$

$$\Rightarrow \boxed{c = \frac{2}{e}}$$

$$\Rightarrow ce = 0$$

$$\Rightarrow \boxed{c = 0}$$

Now, for $c=0$,

②

$$y(x) = (-1)^2 = 1$$

But $y(x)=1$ doesn't satisfy the differential equation and hence

$$c = 2/e.$$

Therefore, the particular solution is:

$$\boxed{y(x) = \left(-1 + \frac{2}{e} e^{e^x}\right)^2}$$

Q7. Solution

Let $y_1(t)$ be the amount of fertilizer in T_1 (in pounds)
and $y_2(t)$ be the amount of fertilizer in T_2 .
(in pounds)

We can model the system with the following differential equations:

$$\textcircled{2} \quad y_1' = \frac{-y_1(16)}{200} + \frac{y_2(4)}{200} + (0)(12)$$

$$y_2' = \frac{y_1(16)}{200} - \frac{y_2(4+12)}{200}$$

$$\Rightarrow \begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} -\frac{16}{200} & \frac{4}{200} \\ \frac{16}{200} & -\frac{16}{200} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

Consider

$$A = \begin{bmatrix} -\frac{16}{200} & \frac{4}{200} \\ \frac{16}{200} & -\frac{16}{200} \end{bmatrix}$$

Now;

$$A - \lambda I = \begin{bmatrix} -\frac{16}{200} - \lambda & \frac{4}{200} \\ \frac{16}{200} & -\frac{16}{200} - \lambda \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-16 - 200\lambda}{200} & \frac{4}{200} \\ \frac{16}{200} & \frac{-16 - 200\lambda}{200} \end{bmatrix}$$

3

$$= \frac{1}{200} \begin{bmatrix} -16 - 200\lambda & 4 \\ 16 & -16 - 200\lambda \end{bmatrix}$$

Now,

$$\det(A - \lambda I) = \left(\frac{1}{200}\right)^2 \det \begin{bmatrix} -16 - 200\lambda & 4 \\ 16 & -16 - 200\lambda \end{bmatrix}$$

$$\text{So, } \det(A - \lambda I) = 0$$

$$\Rightarrow \begin{vmatrix} -16 - 200\lambda & 4 \\ 16 & -16 - 200\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-16-200\lambda)^2 - 64 = 0$$

$$\Rightarrow (-16-200\lambda)^2 = 64$$

$$\Rightarrow -16-200\lambda = \pm \sqrt{64}$$

$$\Rightarrow -16-200\lambda = \pm 8$$

$$\text{So, } -16-200\lambda = 8 \quad ; \quad -16-200\lambda = -8$$

$$\Rightarrow -200\lambda = 24$$

$$\Rightarrow -200\lambda = 8$$

$$\Rightarrow \lambda = \frac{-24}{-200} \quad \begin{matrix} \text{12} \times 2 \\ \text{100} \div 25 \end{matrix}$$

$$\Rightarrow \lambda = \frac{-8}{-200} \quad \begin{matrix} \text{4} \times 2 \\ \text{100} \div 25 \end{matrix}$$

$$\Rightarrow \lambda = -\frac{3}{25}$$

$$\Rightarrow \lambda = -\frac{1}{25}$$

So, $\lambda_1 = -\frac{1}{25}$ and $\lambda_2 = -\frac{3}{25}$ are the eigenvalues of A.

Let v_1 be an eigenvector ^{of A} corresponding to an eigenvalue $\lambda_1 = -\frac{1}{25}$ then

$$A v_1 = \lambda_1 v_1$$

$$\Rightarrow (A - \lambda_1 I) v_1 = 0$$

$$\Rightarrow \left(A + \frac{1}{25} I\right) \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \left(\because \frac{1}{25} = \frac{8}{200}\right)$$

$$\Rightarrow \begin{bmatrix} -\frac{16}{200} + \frac{8}{200} & \frac{4}{200} \\ \frac{16}{200} & -\frac{16}{200} + \frac{8}{200} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow \frac{1}{200} \begin{bmatrix} -8 & 4 \\ 16 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

1.5

$$\Rightarrow -8x + 4y = 0$$

$$\text{and } 16x - 8y = 0$$

$$\text{So, } -8x + 4y = 0$$

$$\Rightarrow -2x + y = 0$$

$$\Rightarrow y = 2x. \quad \text{So, } \begin{pmatrix} x \\ y \end{pmatrix} = t \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Choose $t=1$:

$$v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

This is the eigenvector corresponding to the eigenvalue $\lambda_1 = -\frac{1}{25}$.

For $\lambda_2 = -\frac{3}{25} = -\frac{24}{200}$.

Then

$$A - \lambda_2 I = \begin{pmatrix} -\frac{16}{200} & \frac{4}{200} \\ \frac{16}{200} & -\frac{16}{200} \end{pmatrix} - \begin{pmatrix} -\frac{24}{200} & 0 \\ 0 & -\frac{24}{200} \end{pmatrix}$$

1.5

$$= \begin{pmatrix} \frac{8}{200} & \frac{4}{200} \\ \frac{16}{200} & \frac{8}{200} \end{pmatrix}$$

So,

$$\begin{pmatrix} \frac{8}{200} & \frac{4}{200} \\ \frac{16}{200} & \frac{8}{200} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow 8x + 4y = 0$$

$$\Rightarrow 2x + y = 0$$

$$\Rightarrow y = -2x$$

let $x = t$

$$\text{then } \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} t \\ -2t \end{pmatrix} = t \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\Rightarrow v_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

This is an
eigenvector corresponding to an
eigenvalue $\lambda_2 = -\frac{3}{25}$.



Therefore, the eigenvalues and the eigenvectors of the matrix A are:

$$\lambda_1 = \frac{-3}{25} \quad v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\lambda_2 = \frac{-1}{25} \quad v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The general solution of the ODE system is

$$y = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{-3/25 t} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-1/25 t}$$

At $t=0$, we have

$$y(0) = \begin{bmatrix} 100 \\ 200 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\textcircled{2} \Rightarrow \begin{aligned} c_1 + c_2 &= 100 \\ -2c_1 + 2c_2 &= 200 \end{aligned} \quad \boxed{\times 2}$$

So,

$$2c_1 + 2c_2 = 200$$

$$\begin{array}{r} -2c_1 + 2c_2 = 200 \\ \hline 4c_2 = 400 \end{array}$$

$$4c_2 = 400 \Rightarrow c_2 = 100 \text{ and so}$$

$$c_1 + c_2 = 100 \Rightarrow c_1 = 0$$

So, the solution is

$$y = 100 \begin{bmatrix} 1 \\ 2 \end{bmatrix} e^{-t/25}.$$

THE

END